

Introduction

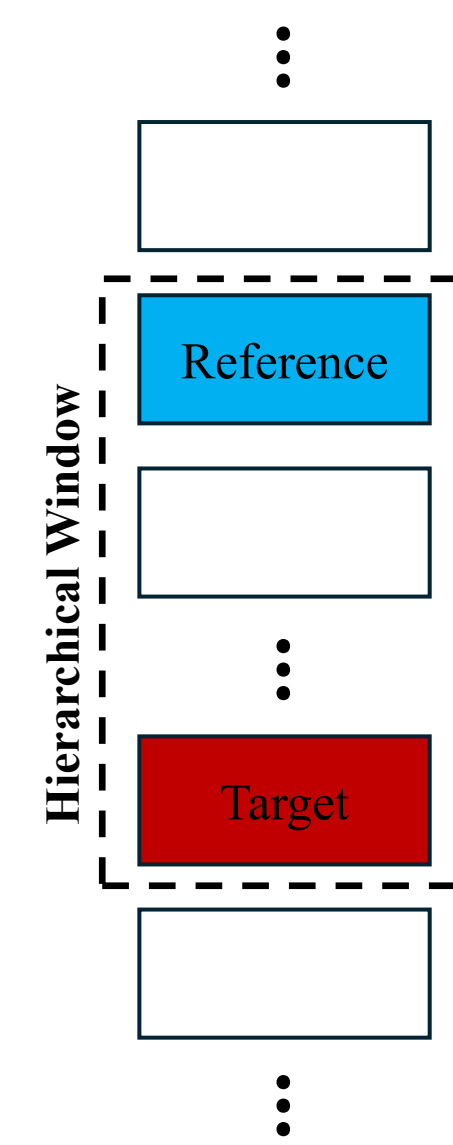
- The visual system hierarchically decomposes multiple motion vectors into shared and relative components (Johansson, 1973).
- Overestimation of the relative motion component produces ‘induced motion’ (Duncker, 1938).
- Human visual motion perception exploits hierarchical latent motion structure through Bayesian-like inference (Gershman et al., 2016; Bill et al., 2020).

Question:

- Can the visual system decompose a complex hierarchical motion structure consisting of more than two layers?

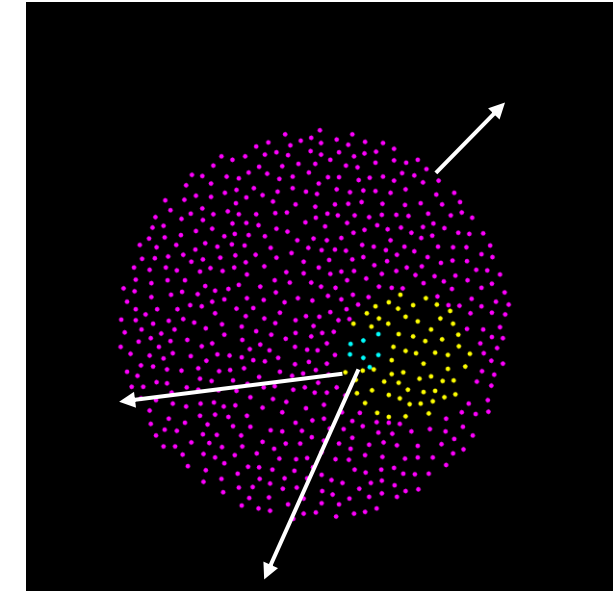
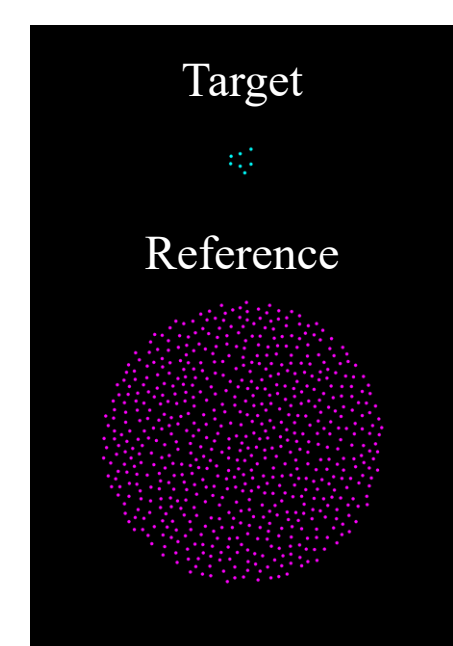
Prediction:

- If ‘yes’, we expect that the judgment of Target motion relative to Reference motion will be affected by other irrelevant components (e.g., induced motion) if and only if they are inside the hierarchical window bounded by the Target and Reference.

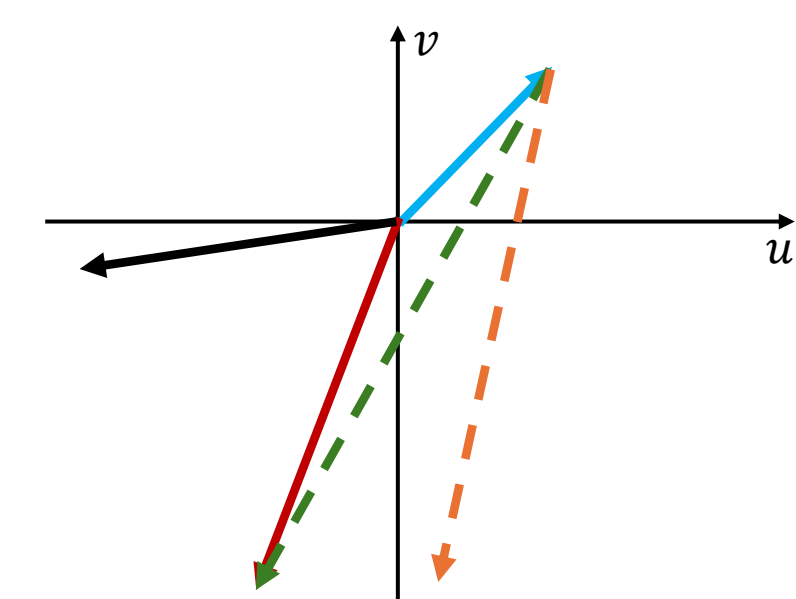
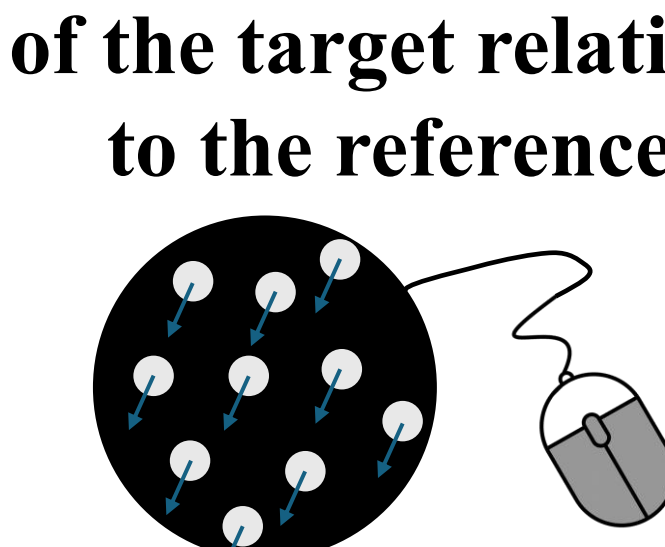


Method

Definition of Target & Reference	Stimulus presentation for 1s, 60 fps, 3 repeats	Observer's response: reproduction of the motion
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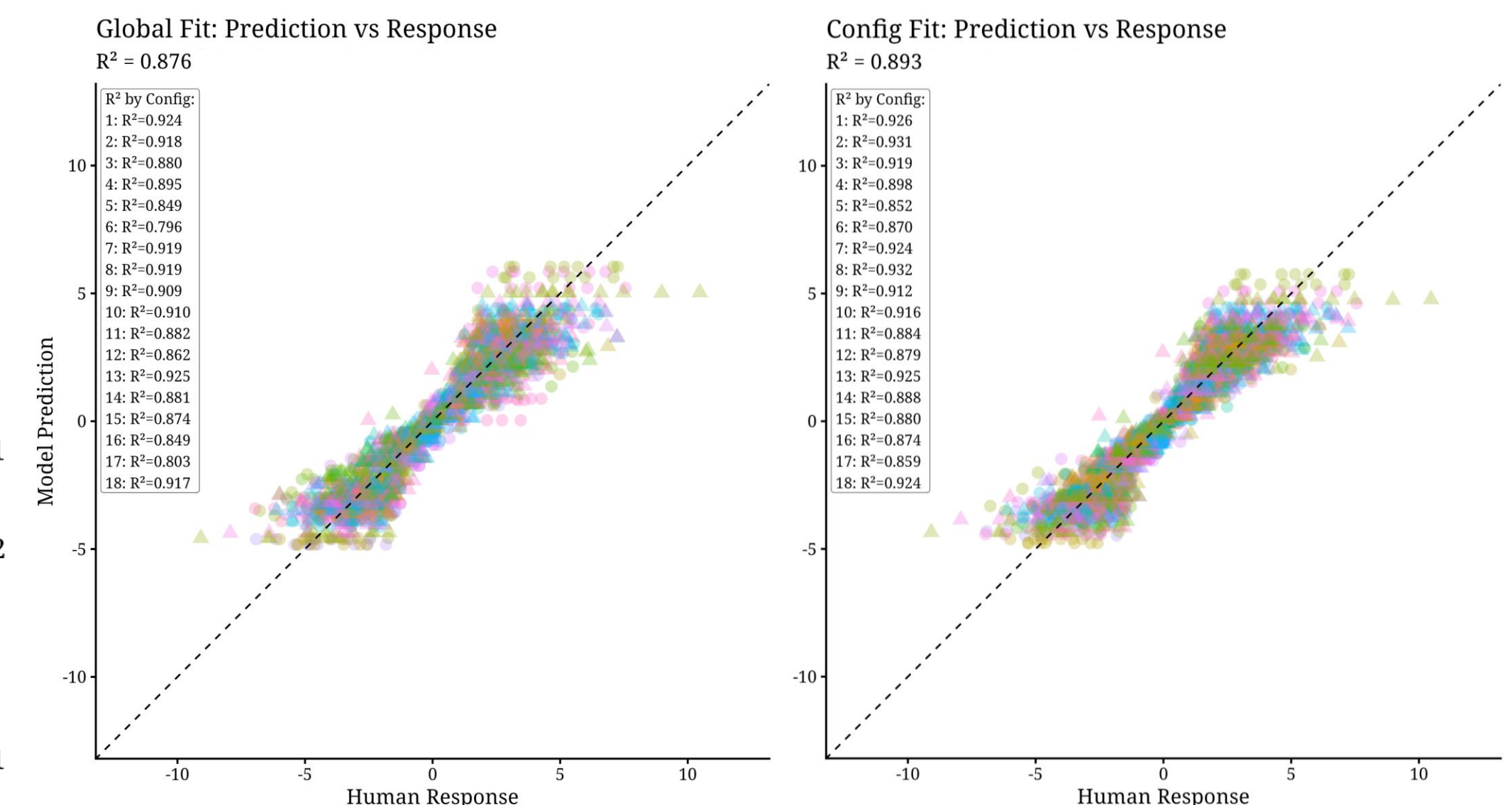
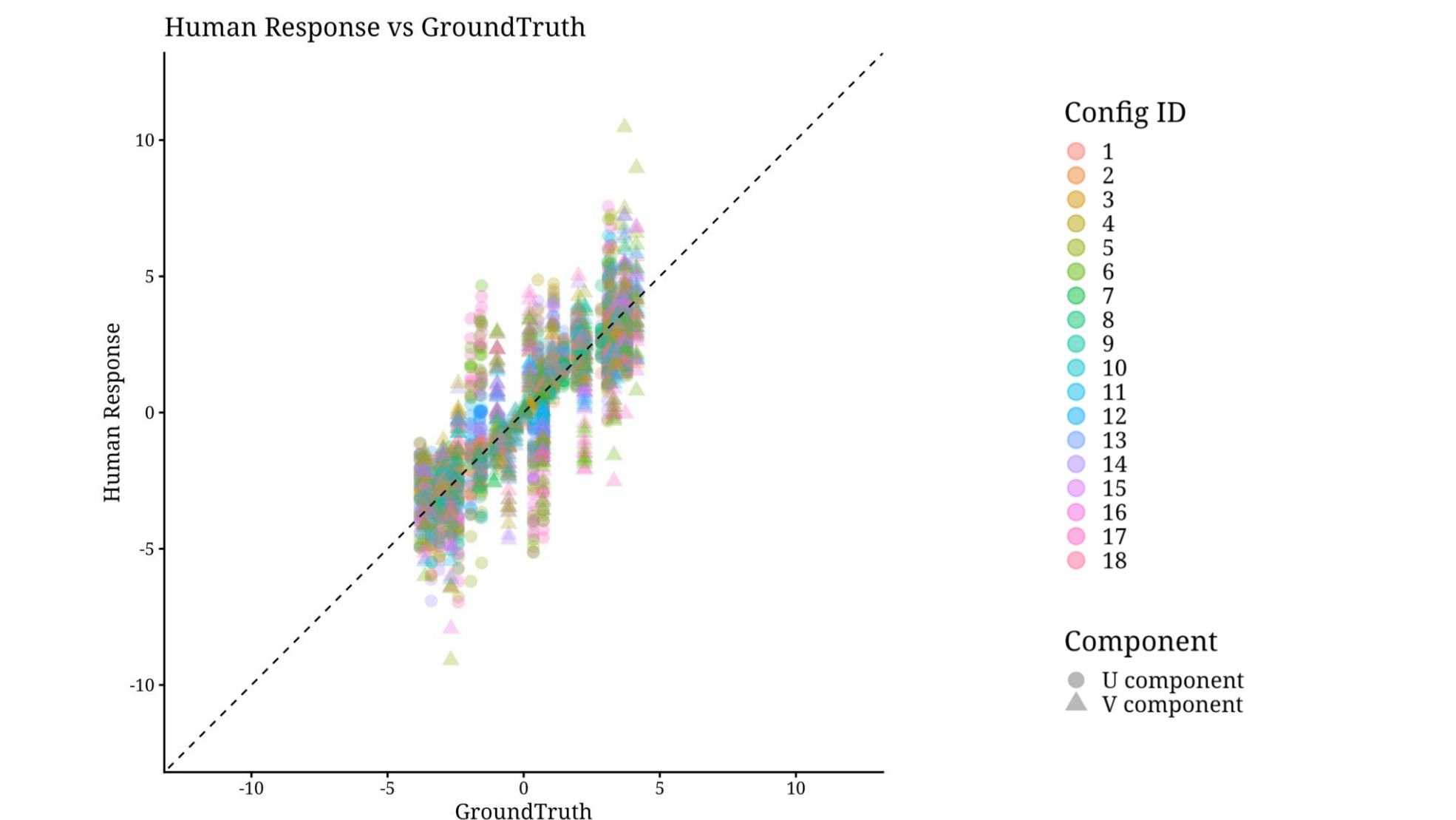
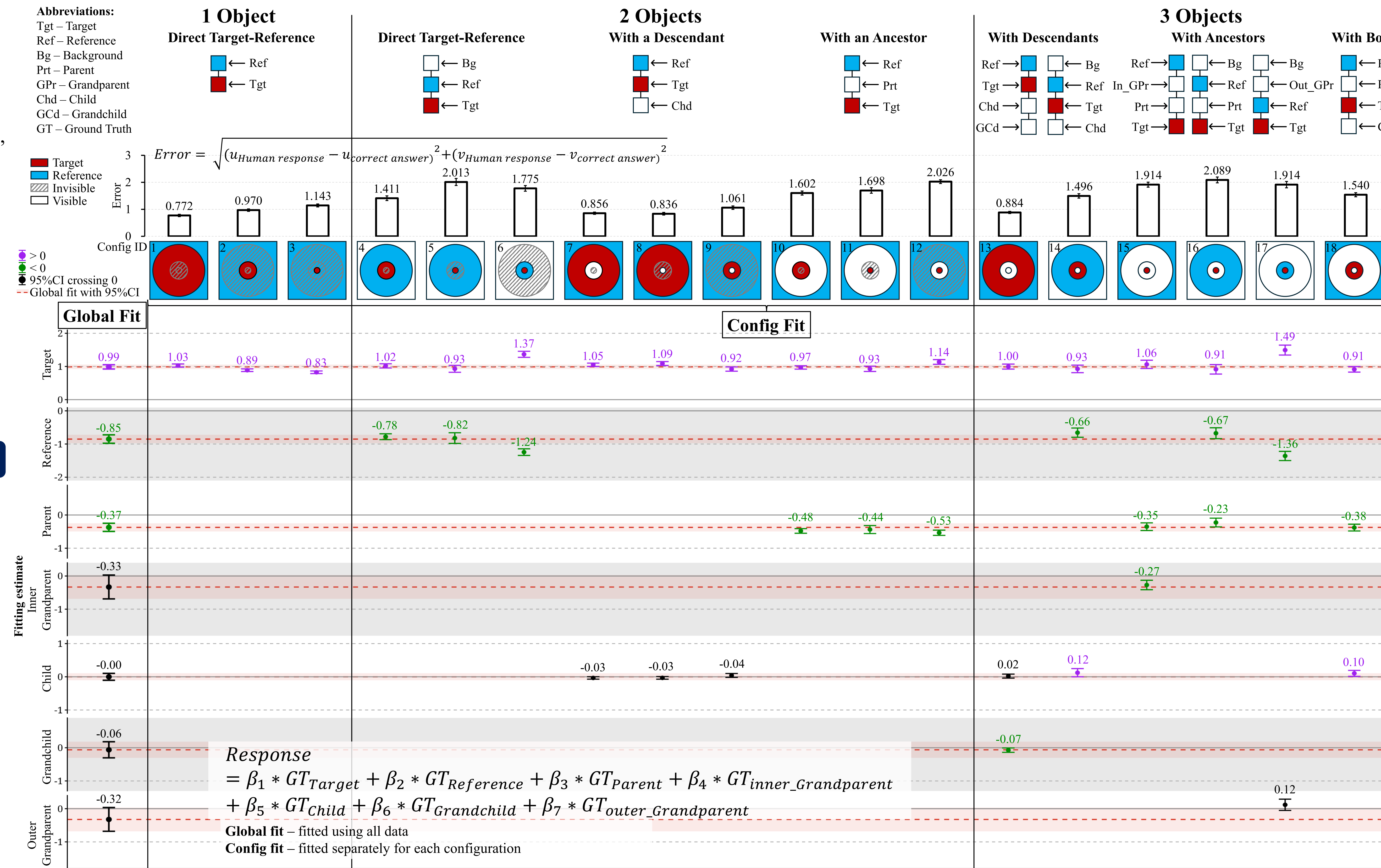
The stimulus consists of 1, 2, or all 3 objects



- **Target**
- **Reference**
- **Irrelevant motion (e.g., Parent)**
- → **Correct Answer**
- → **Human Response**

$$\text{Correct Answer} = GT_{\text{Target}} - GT_{\text{Reference}}$$

Results



Conclusions

- Induced-motion-like interactions occurred only for components within the hierarchical window bounded by the Target and Reference, whereas those outside the window did not contribute.
- Not only a single contextual component (classic induced motion), but also multiple nested components influenced motion perception.
- These findings support the idea that the visual system can decompose motion hierarchically across multiple components.